

BFMC 2026 - Status Report 1

Date: Monday, December 22nd, 2025

Prepared by: Pothole Avoidance

Reporting Period: November 28th to December 22nd

Overview of Project Approach:

For the 2026 edition of BFMC, the goal for Pothole Avoidance is to first rebuild and validate the architecture elaborated for the 2025 edition, teaching new and less experienced team members to use it. This means reacquiring the hardware from last year, assembling it and running integration tests, then comparing performance to that achieved last year. Once parity with last year's capabilities is achieved, improvement of the existing architecture by testing new ideas (such as a downward facing optical sensor for odometry) can be undertaken.

Current Phase:

Currently, the team is still assembling the components required to recreate the 2025 Pothole Avoidance vehicle, with new members of the team learning the underlying concepts that will be required for them to operate it.

Task Status:

Architecture Component Sourcing:

- Kit Car received and confirmed to function (Demo run)
- Realsense and Jetson purchased and received
- STM32 (embedded platform) and Magnetic Encoder leftover from previous years
- PETG filament required for 3D printed mounts has been purchased, and printing of mounting components is in progress

Next Steps:

1. Connecting all electronic components temporarily to confirm wiring and correct function.
2. Completing vehicle assembly.
3. Steering and speed calibration tests on assembled vehicle
4. Integration test
5. Determining key areas of architecture weakness.